

Identifying Variables — Set B — Answer Key

Level 2 · Ages 9-11 · Answers in bold. Accept any answer that means the same.

Independent variable — the one thing you change on purpose.

Dependent variable — what you measure to see what happened.

Controlled variables — things you keep the same so the test is fair.

- ① Lucy puts 1, 2, 3 and then 4 sheets of paper between a magnet and a pile of paper clips. Each time she counts how many clips the magnet picks up. She uses the same magnet and the same paper.

Independent variable: The number of sheets of paper (1-4); more sheets = more distance.

Dependent variable: The number of paper clips the magnet picks up.

Two controlled variables: The same magnet and the same kind of paper. (Also accept: the same paper clips.)

- ② Ben adds one teaspoon of sugar to 100 mL of water at 20 °C, 30 °C and 40 °C, stirring steadily the whole time. He times how long the sugar takes to dissolve.

Independent variable: The temperature of the water (20, 30 and 40 °C).

Dependent variable: The time the sugar takes to dissolve (seconds).

Two controlled variables: Any two: one teaspoon of sugar; 100 mL of water; the same steady stirring.

- ③ Zara puts three identical bean plants in a bright window, a shady corner and a dark cupboard. Each plant gets 50 mL of water every day. After two weeks she counts the leaves on each plant.

Independent variable: The amount of light (where each plant is placed).

Dependent variable: The number of leaves after two weeks.

Two controlled variables: Any two: 50 mL of water a day; identical plants; two weeks.